

Breaking Bad News[†]

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We study how information disclosure shapes social learning about a potentially harmful product. Increased transparency helps early agents avoid harm, which may undermine learning by later agents. Despite this conflict of interest, we show that full transparency is uniquely optimal for all agents when they learn only by observing neighbors' harm. We investigate whether full transparency about harm continues to benefit all agents when they also learn from additional, imperfect signals. (JEL D18, D82, D83, L15)

When a new drug enters the market, many patients will benefit, but there is a risk of severe side effects. Should regulators require doctors to immediately report all adverse outcomes to the entire medical community or allow limited, decentralized learning to unfold before mandating disclosure? Similar dilemmas arise across a wide range of domains: Freelance platforms decide whether to share client reviews about the competence of workers, industry watchdogs can reveal whether a new gadget harms customers, and government agencies may disclose whether a financial product is fraudulent. In each case, there is a trade-off between exploration and the exploitation of information: Early decisions produce information that benefits others but may come at a steep personal cost. Should such information be made widely available?¹

This paper studies how and when bad news should be shared. We model a sequence of agents who can consume a product of unknown quality; the product may harm consumers if its quality is low, but not if quality is high. Agents observe the outcomes of some earlier-moving “neighbors”; we interpret the resulting information structure as a network. We ask: What network design maximizes overall welfare?

This enterprise faces two theoretical challenges. First, it is complicated to solve for Bayesian equilibrium on a given network, let alone solve for the optimal network. Second, adding links can make one agent better off and others worse off.

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¹There is a large empirical literature studying the diffusion of information, including case studies on the above markets: doctors learning about new drugs (e.g., Coleman, Katz, and Menzel 1957), employers learning about freelancers (e.g., Pallais 2014), consumers learning about a new gadget (e.g., Goolsbee and Klenow 2002), and entrepreneurs learning about a new financial product (e.g., Banerjee et al. 2013).

Nevertheless, we will see that the optimal network is remarkably simple: full transparency.

When choosing whether to consume the product, agents ignore the positive informational externality of their actions on their successors. The optimal information policy must therefore account for both the benefits of exploiting current social information and the benefits of exploration. For example, in a good news model where “successes” prove that quality is high, Che and Hörner (2018) show that a forward-looking utilitarian designer mixes in fake successes to stretch the good signals and increase exploration.²

In our setting with bad news learning,³ an agent’s learning and welfare can be summarized by her chance of seeing a neighbor harmed. This suggests a conflict of interest over the network since her neighbors use their own social information to avoid being harmed. Despite this tension, Theorem 1 shows that every type of every agent prefers the complete network to all other networks. While each agent uses her neighbors as guinea pigs, full transparency optimally targets their exploration toward the states where the agent is uninformed. Thus, while agents differ in their preferences over adding a single link, they all agree on the optimal network.

This result provides a logical foundation for the transparency policies that prioritize monitoring and sharing “bad news events” over similar “good news events.” Such policies are often seen in practice: The FDA requires that drug companies report adverse health events, the CPSC requires that manufacturers report unsafe products, and the SEC requires that customers be notified of fraud.

Our results also inform policy when full transparency is not feasible. A transitive network, in which each agent sees everything her neighbors see, Pareto dominates all of its subnetworks. A consequence is that, even when information arises with delay, permanent public warnings lead to Pareto improvements, whereas temporary warnings do not. We also establish a sense in which transparency is complementary, meaning that the returns to informing some people are higher when other people are already informed.

We end the paper by asking whether public warnings about perfect bad news outcomes also constitute a Pareto improvement in models with additional outcomes that provide inconclusive evidence about low or high quality (e.g., when agents learn from their predecessors’ actions). A simple public warning triggered by bad news can make some agents worse off. However, an information designer who can directly communicate with the agents and control their information can fix this problem and render such warnings a Pareto improvement using a robust and detail-free mechanism. Following the information design literature, we discuss how these insights depend on the designer’s information regarding agents’ private types.

²Relatedly, Sgroi (2002) and Kremer, Mansour, and Perry (2014) show that full transparency is suboptimal in models with, respectively, observable actions and imperfect outcome signals. Che and Hörner (2018) also characterize the welfare-maximizing information design in a model with both good news and bad news signals; this mechanism converges to full transparency as the intensity of good news vanishes. In the limit case of pure bad news, our paper shows more strongly that full transparency is Pareto dominant for a rich set of types.

³Throughout, we use “bad news” as shorthand for perfect bad news that proves that quality is low. We use “imperfect bad news” when referring to noisy signals.

Literature.—We contribute to a large literature studying social learning starting with Bikhchandani, Hirshleifer, and Welch (1992). Our network model follows Acemoglu et al. (2011) in that agents observe information from their neighbors in addition to a private signal. The key difference is that our agents see their neighbors' outcomes instead of their actions, as in Wolitzky (2018) and Board and Meyer-ter-Vehn (2021), and that observing a neighbor harmed constitutes perfect bad news. This allows us to Pareto rank networks.

There is also a large literature on information design (e.g., Bergemann and Morris 2019) that includes models where communication is dynamic (e.g., Ely 2017) or is constrained by a network (e.g., Galperti and Perego 2025). The typical paper in this literature characterizes the optimal information policy for a particular game. In contrast, we are interested in whether a particularly simple policy—fully disclosing bad news—is Pareto improving across a wide range of environments.

More closely related to our result, Knoepfle and Salmi (2024) show that an information designer would release bad news signals without delay in a game of endogenous timing. Our setting with exogenous timing is simpler, allowing us to study a broader range of individualized information policies.⁴

I. Model

Players and Actions.—Agents $\{1, \dots, T\}$ with generic members $r < s < t$ (Ruby, Sven, Tara) consider a new product with idiosyncratic, latent outcomes $\varphi_t \in \{b, n\}$, “bad/harm” and “null,” connected by common quality $q \in \{L, H\}$. At time t , agent t decides (once and for all) whether to consume the product, $c_t \in \{0, 1\}$. Her realized outcome ϕ_t equals her latent outcome φ_t if she consumes, $c_t = 1$, and n otherwise.

Information.—Tara observes the outcomes of agents $G_t \subseteq \{1, \dots, t-1\}$ and a finite type θ_t that shapes her preferences and beliefs. We interpret $\{G_t\}$ as a directed network; when $s \in G_t$, we write $s \leftarrow t$ and call s a *neighbor* of t . We write $\bar{G}$ for the complete network, $\bar{G}_t = \{1, \dots, t-1\}$.

Payoffs and Beliefs.—Consuming the product yields utility $u(\theta_t, \phi_t)$; not consuming yields 0. Agents dislike harm:

$$(1) \quad u(\theta_t, b) < u(\theta_t, n).$$

Formally, the finite state space comprises quality, latent outcomes, and types $\Omega = \{L, H\} \times \{b, n\}^T \times \Theta$. When Tara's belief $\Pr_{\theta_t}(q, \varphi, \theta_{-t})$ assigns probability one to an event, we say she “knows” the event.

Bad News Learning.—High-quality products cannot cause harm; $\Pr_{\theta_t}(\varphi_s = b | H) = 0$ for all θ_t, s . Moreover, quality is a sufficient statistic for

⁴Indeed, with endogenous timing, asymmetric policies that reveal A's outcome to B (but not vice versa) raise utilitarian welfare by inducing A to move immediately, which benefits B.

individual harm; that is, conditional beliefs $\Pr_{\theta_t}(\varphi_t | q, \varphi_{-t}, \theta_{-t})$ are independent of others' outcomes and types $(\varphi_{-t}, \theta_{-t})$.

Solution Concept.—We assume (generically) that all types have strict consumption preferences. Proceeding by induction over t , we solve for the unique rationalizable strategy profile, and hence the unique (Bayesian–Nash) equilibrium of this game. Given this unique equilibrium, we compare networks according to the interim Pareto ranking of all types θ_t of all agents t .

Remarks.—Our key assumption is that the high-quality product harms nobody, so harm constitutes perfect bad news, proving that quality is low. Thus, Tara only faces two relevant classes of histories: the empty history where she observes no harm and the histories where she observes harm, in which she concludes that quality is low. This distinguishes our model from traditional social learning models where agents learn from actions rather than outcomes. We discuss such models in Section IIB.

Our specification of types $\theta = (\theta_1, \dots, \theta_T)$ is very general. Tara's private type θ_t may affect her personal tastes $u(\theta_t, \phi_t)$ (as in Goeree, Palfrey, and Rogers 2006); private signals about quality $p(\theta_t) := \Pr_{\theta_t}(H)$ (as in Bikhchandani, Hirshleifer, and Welch 1992); and the probability of idiosyncratic harm $\alpha(\theta_t) := \Pr_{\theta_t}(\varphi_t = b | L)$. The model does not require a common prior over Ω and allows for arbitrary correlations. Our results easily extend to settings with timing uncertainty, where t moves at some time $\tau(\theta_t)$, and to random networks.

Our positive results, that some network G (weakly) dominates some network G' , hold for every state space Ω . Strict dominance and counterexamples—for example, Example 1—obtain by constructing types θ_t who do not prefer G over G' ; we assume throughout that our finite state space Ω is rich enough to include these types.

II. Optimal Network

We now show that the fully transparent, complete network $\bar{G}$ Pareto dominates all other networks. In practice, $\bar{G}$ corresponds to a public warning when anyone is harmed.

The conditional probability that Tara observes harm in the unique equilibrium,

$$(2) \quad y(\theta_t) = \Pr_{\theta_t}(\phi_s = b \text{ for some } s \in G_t | L),$$

is a sufficient statistic for her social information about quality (which in turn is a sufficient statistic for her latent outcome φ_t). The higher is $y(\theta_t)$, the Blackwell higher is θ_t 's information and hence welfare. This formalizes the notion that t uses her neighbors as guinea pigs.

When Tara observes no harm, she updates her quality belief to

$$(3) \quad p^n(\theta_t, y(\theta_t)) := \frac{p(\theta_t)}{1 - (1 - p(\theta_t))y(\theta_t)}$$

and her no-harm belief to $1 - (1 - p^n(\theta_t, y(\theta_t)))\alpha(\theta_t)$. She consumes if the latter exceeds the threshold $x^*(\theta_t)$ that uniquely solves her indifference condition



FIGURE 1. NETWORKS FROM EXAMPLES 1 AND 2

$xu(\theta_r, n) + (1 - x)u(\theta_r, b) = 0$.⁵ When Tara observes harm, her quality belief jumps to 0, and she consumes if her no-harm belief $1 - \alpha(\theta_t)$ exceeds $x^*(\theta_t)$.

We first show that “indirect links” can make agents worse off.

EXAMPLE 1: *Start with a network where Tara observes Sven, and add a link from Sven to Ruby (see Figure 1, left). Clearly, the new link benefits Sven by giving him more information to exploit. But its effect on Sven’s exploration and Tara’s learning is ambiguous: If Ruby is harmed, Sven grows pessimistic and less likely to consume, which is bad for Tara; if Ruby is unharmed, Sven grows optimistic and more likely to consume, which is good for Tara.*

On average, the new link may make Tara worse off. Consider the following parametric example: Types capture private consumption costs $\theta_t = k_t \sim U[0, 1]$, independent across agents and independent of quality q ; the common quality belief is $p(\theta_t) \equiv p$ and low quality harms all consumers, $\alpha(\theta_t) = 1$.⁶ Assume that utility $u(\theta_r, b) = -k_t < 1 - k_t = u(\theta_r, n)$, inducing cutoff $x^(\theta_t) = k_t$. Sven consumes the low-quality product and is harmed if his random posterior P_s (which equals $p^n(\theta_s, y(\theta_s))$ with probability $1 - y(\theta_s)$ and 0 otherwise) exceeds k_s . All told, Tara’s conditional probability of seeing Sven harmed is $y(\theta_t) = \Pr_{\theta_t}[P_s \geq k_s | L] = E_{\theta_t}[P_s | L]$. Any rise in Sven’s information, such as the link to Ruby, helps Sven avoid low-quality products and lowers $E_{\theta_t}[P_s | L]$; this lowers Tara’s social learning and welfare.⁷*

The next example shows how full transparency resolves Tara and Sven’s conflict of interest.

EXAMPLE 2: *Suppose that Tara initially observes Ruby as well as Sven, and consider again the effect of adding a link from Sven to Ruby (see Figure 1, right). Since Tara observes Ruby directly and seeing one neighbor harmed is proof of low quality, she cares about learning from Sven only in the event that Ruby is unharmed. And in this event, the link $r \leftarrow s$ makes Sven unambiguously more optimistic, raising his chance of consumption and thereby Tara’s social learning and welfare.*

Full transparency for Sven is thus optimal for Sven and Tara. More broadly, Tara wants Sven to observe any proof of low quality that she observes herself, since this

⁵The solution $x^*(\theta_t) \in \mathbb{R}$ is unique by (1). If $u(\theta_r, b) < 0 < u(\theta_r, n)$, we have $x^*(\theta_t) \in (0, 1)$.

⁶Formally, we interpret this continuous-type space as the limit of finite-type spaces in our model.

⁷Looking across agents, Tara’s loss from Sven’s link to Ruby is balanced by Sven’s gain, so the effect on utilitarian welfare is ambiguous. But if Tara has many clones who all observe the same Sven, then the link $r \leftarrow s$ reduces utilitarian welfare.

makes him the most effective guinea pig by directing his exploration to the event where it maximizes Tara's learning.⁸

To generalize Examples 1 and 2, call network G *transitive* if whenever Tara indirectly observes Ruby via Sven, then Tara also observes Ruby directly; formally, $r \in G_s$ and $s \in G_t$ imply that $r \in G_t$. For instance, the complete network $\bar{G}$ is transitive.

THEOREM 1: *A network G Pareto dominates all of its subnetworks $G' \subseteq G$ if and only if G is transitive. Thus, the complete network $\bar{G}$ Pareto dominates all other networks.*

PROOF:

Define the set of states $\omega = (q, \varphi, \theta)$, where t observes harm, or equivalently, a neighbor of t with a latent harm outcome consumes the low-quality product

$$\begin{aligned}\Omega_t &= \{\omega : \phi_s = b \text{ for some } s \in G_t\} \\ &= \{(L, \varphi, \theta) : c_s = 1, \varphi_s = b \text{ for some } s \in G_t\}.\end{aligned}$$

By definition, $y(\theta_t) = \Pr_{\theta_t}(\Omega_t | L)$, so θ_t 's information and hence welfare rise in Ω_t . Writing $\Omega'_t, y'(\theta_t), c'_t$ for the respective variables in network G' , we now show that $\Omega'_t \subseteq \Omega_t$ for all $G' \subseteq G$ if and only if G is transitive.

That transitivity is necessary essentially follows from Example 1. Suppose that G is not transitive, so there are agents r, s, t with $s \in G_t$ and $r \in G_s$, but $r \notin G_t$. Consider the subnetwork $G' \subset G$ with $s \in G'_t$ and $r \notin G'_s$, and a state ω where (i) Ruby tries and is harmed, (ii) Sven tries unless he knows that quality is low, and (iii) no other agent tries. So constructed, $\omega \notin \Omega_t$ but $\omega \in \Omega'_t$; Tara prefers G' over G if she thinks ω is likely.

We argue sufficiency by induction. The anchor $\Omega_1 = \Omega'_1 = \emptyset$ is trivial. Assume that $\Omega_s \supseteq \Omega'_s$ for all $s < t$, and fix $\omega = (L, \varphi, \theta) \in \Omega'_t$, so Tara's neighbor Sven with type θ_s consumes and is harmed in G' . Now consider network G ; we wish to show that $\omega \in \Omega_t$. If some neighbor of Sven's, Ruby, is harmed, then by transitivity Tara also observes that Ruby is harmed. If none of Sven's neighbors are harmed, then the induction hypothesis renders Sven more optimistic in G , $p^n(\theta_s, y(\theta_s)) \geq p^n(\theta_s, y'(\theta_s))$ since

$$y(\theta_s) = \Pr_{\theta_s}(\Omega_s | L) \geq \Pr_{\theta_s}(\Omega'_s | L) = y'(\theta_s).$$

Thus, Sven consumes and is harmed in G . In either case, $\omega \in \Omega_t$, concluding the induction step. ■

⁸ While Tara and Sven agree on full transparency, they would not agree on other policies. Indeed, if Tara could force guinea pig Sven to consume the product, as in Smith, Sørensen, and Tian (2021), she would do so, which is clearly not in Sven's interest. It is thus important that the only design variable is the social information agents learn from one another.

Since observing harm perfectly reveals low quality and nothing remains to be learned, one might wonder why the case for full transparency even requires a proof. And yet, Che and Hörner (2018) show that signals that perfectly reveal high quality may lower ex ante welfare: Such information makes agents more pessimistic in the absence of good news signals and reduces socially beneficial experimentation. In contrast, the absence of bad news signals makes agents more optimistic. Indeed, in the good news version of our model,⁹ the optimal network for all types θ_i of Tara is the *t-star*, where she observes all other agents, and other agents observe nobody.¹⁰ This, of course, is the worst network for all other agents, meaning that there is maximal conflict over the network.¹¹

Transparency is optimal for bad news signals but not for good news signals. How about other signals? By continuity and strict preferences, Theorem 1 extends to mostly harmless high-quality products that cause harm with some small probability ϵ . But for larger values of ϵ , some types require two harm signals to change their actions, so the complete network is no longer optimal; for example, the correlating link $r \leftarrow s$ in Example 2 may dissuade Sven from consumption when Ruby is harmed, depriving Tara of a second harm signal.

A. Implications of Theorem 1

Theorem 1 shows that full transparency is Pareto dominant; it also provides sufficient conditions for Pareto-improving policies that fall short of full transparency. First, consider an organization in which managers observe the outcomes of the subordinates who report to them. Our directed networks can describe traditional hierarchies and matrix organizations, where workers have multiple managers, but rule out two-way reporting. In such a setting, adding new reporting links Pareto improves if the resulting network allows supervisors to observe the outcomes of their indirect as well as their direct reports.¹²

Second, suppose that bad news is subject to delay (e.g., it takes time for a drug's side effects to materialize), so Sven's outcome can only be observed after $\tau(s) \geq s$. The maximally transparent network G^τ with $G^\tau = \{s : \tau(s) < t\}$ is transitive, so a public warning is Pareto dominant. Conversely, suppose that the public memory is limited, and Sven's outcome fades after $\tau(s) \geq s$; the maximally transparent

⁹The good news model flips inequality (1), swaps “high” and “low” quality, and replaces “bad/harm” with “good/success.”

¹⁰*Proof:* Fix a state ω and let $s < t$ be the first agent who succeeds in some network G . Since s has not observed a success in G , she is less optimistic than in the *t-star*. Thus, she also tries and succeeds in the *t-star*. All told, t observes more successes and hence has Blackwell-superior information in the *t-star*.

¹¹In contrast to our deterministic star network, Che and Hörner (2018) allow for stochastic recommendations. Their utilitarian information designer improves on the star network by merging some “no news” states with the “good news” states until the agent is indifferent. With bad news learning, full transparency also Pareto dominates stochastic mechanisms M where Tara's message conditions on past outcomes and a noise term $m(\phi_1, \dots, \phi_{t-1}, \epsilon_t)$. The proof adapts the proof of Theorem 1. Let Ω_t^M be the set of states where some $s < t$ is harmed for some signal realization in M . Assume inductively that $\Omega_t^M \subseteq \Omega_t^G$. Since M cannot distinguish states outside Ω_t^M , the “null” message under full transparency renders Tara more eager to try than any message in M , inducing Tara to try more, concluding the induction step.

¹²Theorem 1 states that G Pareto dominates $G' \subset G$ if G is transitive (i.e., if managers see the outcomes of all indirect reports). However, transitivity of G is not necessary to dominate any particular G' . Indeed, the proof of Theorem 1 shows that adding the link $r \leftarrow s$ in Example 2 remains Pareto improving if additionally Ruby (but not Sven or Tara) sees all of Ruby's predecessors in G' .

network $\hat{G}^\tau$ with $\hat{G}_t^\tau = \{s : \tau(s) \geq t\}$ is not transitive, so maximal transparency need not be optimal as in Example 1. This illustrates how temporary warnings can backfire and makes the case that warnings should be permanent.

Third, Theorem 1 can inform policy when budget constraints preclude full transparency. Formally, for any subset $J \subset I := \{1, \dots, T\}$, let $U^J(\theta_t)$ be type θ_t 's expected utility in the network where agents $t \in J$ observe all predecessors and $t \notin J$ none. Transparency is complementary in that

$$(4) \quad U^I(\theta_t) - U^{I \setminus J}(\theta_t) \geq U^J(\theta_t) - U^\emptyset(\theta_t)$$

for all θ_t .¹³ For instance, with three agents $I = \{r, s, t\}$ and $J = \{t\}$, equation (4) reduces to Example 2: Observing Sven and Ruby is more valuable for Tara when Sven also observes Ruby; intuitively, Sven's link to Ruby makes him a more effective guinea pig by targeting his exploration to the state where Ruby has succeeded. Thus, if agents in two cities are learning about the quality of a city-specific product (e.g., whether drugs are laced with fentanyl), it is better to fully inform one city than to partially inform both. Similarly, if information is better in one of the cities (e.g., due to local nonprofits sharing information), then the designer should, perhaps counterintuitively, focus on providing information to the city that is already better informed.

While Theorem 1 concerns agents' welfare, it implies that full transparency is also optimal for high-quality sellers since $\bar{G}$ maximizes demand $\sum_t c_t$ in all high-quality states ω .¹⁴ The effect on the average seller depends on parameters: With the uniform cost types from Example 1, Tara's expected consumption equals the prior probability of high-quality $E[P_t] = p$, so unconditional expected demand equals $E[\sum_t c_t] = Tp$, independent of the network. For these types, the complete network then minimizes expected demand for low-quality sellers, as well as maximizing it for high-quality sellers.¹⁵

B. Transparency with General Outcomes

The key assumption behind Theorem 1 is that observing harm is both perfectly informative of low quality and the only learning channel. In practice, doctors may observe both perfect bad news outcomes (e.g., birth defects from thalidomide) and other, imperfectly informative outcomes (e.g., colleagues' prescription choices and the drug's effectiveness on patients). We have seen that full transparency of all outcomes need not be optimal in such settings (e.g., for good news signals). But it seems intuitive that the FDA still wants to publicly announce adverse events that prove low quality. In this subsection, we show that simply revealing such events may

¹³ *Proof:* Agents $t \in J$ start from the same fully uninformed baseline, $U^{I \setminus J}(\theta_t) = U^\emptyset(\theta_t)$, and by Theorem 1 are best off with full transparency, $U^I(\theta_t) \geq U^J(\theta_t)$. Agents $t \notin J$ have nothing to gain if they are fully uninformed, $U^J(\theta_t) = U^\emptyset(\theta_t)$, but by Theorem 1 benefit from J 's transparency when they themselves have transparency, $U^I(\theta_t) \geq U^{I \setminus J}(\theta_t)$.

¹⁴ *Proof:* The proof of Theorem 1 shows that whenever type θ_t consumes in network G , she also consumes in the complete network $\bar{G}$ when harm has not yet occurred, as is the case for $q = H$.

¹⁵ This has interesting implications for settings where the seller knows the quality of its good and designs the network G before the game. In such a signaling game, the natural equilibrium is for the high-quality seller to signal high quality by choosing $\bar{G}$, forcing the low-quality seller to pool and also choose full transparency.

introduce trade-offs, but an information designer who can directly communicate with the agents and control their information can achieve a Pareto improvement.

To study this, we consider general, finite outcomes $\varphi_t \in \Phi_t$ that include b and n from the baseline model. Agents are connected via a network G , with each agent observing the outcomes of neighbors, $\phi_{G_t} = (\phi_s)_{s \in G_t}$. As in the baseline, actual outcomes are $\phi_t = \varphi_t$ if $c_t = 1$ and $\phi_t = n$ if $c_t = 0$, and high quality prevents harm $\Pr_{\theta_t}(\varphi_s = b | H) = 0$ for all s, t ; thus, observing bad news perfectly reveals $q = L$, which lowers consumption utility

$$E_{\theta_t}[u(\theta_t, \varphi_t) | L] < E_{\theta_t}[u(\theta_t, \varphi_t) | H],$$

generalizing (1). A salient special case is observable actions $\Phi = \{b, n, cn\}$ for harm, nonconsumption, and consumption with no harm.

First, consider a public warning G^+ that augments a given network G by publicly revealing harm. Formally, under G^+ , agent t observes outcomes ϕ_s^+ of all neighbors and additionally a binary harm indicator ζ_t^+ that equals b if and only if any predecessor $s < t$ has been harmed, $\phi_s^+ = b$; else $\zeta_t^+ = n$. This intuitive transparency policy fails to benefit all types.

EXAMPLE 3: *Network G has two links, $r \leftarrow s \leftarrow t$, as in Example 1. Additionally, there are some prior, unconnected agents that may have experienced harm. Suppose that neighbors' actions are observable, $\Phi = \{b, n, cn\}$. Consider a type θ_t who knows that:*

- (i) *Ruby's latent outcome is $\varphi_r = cn$. In the original network G , Ruby does not consume, $\phi_r = n$. In G^+ , when $\zeta_r^+ = n$ renders her more optimistic, her consumption and hence her outcome $\phi_r^+ \in \{n, cn\}$ depends on her type θ_r , which carries quality information (as in Bikhchandani, Hirshleifer, and Welch 1992).*
- (ii) *Sven's latent outcome perfectly reveals quality: $\phi_s = b$ when $q = L$, and $\phi_s = cn$ when $q = H$. In the original network G , Sven consumes when Ruby abstains, $\phi_r = n$. In G^+ , he does not consume when Ruby abstains, $\phi_r^+ = n$ (which is now negative news about Ruby's type, and hence quality).*

Type θ_t learns quality perfectly in G but imperfectly in G^+ . Hence, the public warning makes θ_t worse off.¹⁶

The problem with the public warning in Example 3 is that while it fixes the observation network for outcomes other than harm, it alters Ruby's and Sven's actions and outcomes. Specifically, in the example, it renders Sven more pessimistic and dissuades him from trying when Ruby doesn't try. We now propose an adjusted public

¹⁶Example 3 shows that in most networks, the public warning does not raise the welfare of all types of all agents. This is clearly an exacting welfare measure, and it seems likely that for many natural-type spaces and networks G , the harm broadcast G^+ does raise utilitarian welfare.

warning that benefits all agents. To do so, we introduce an information designer who observes all past outcomes and can control agents' information.

Consider an *adjusted public warning* G^* . As before, the designer tells Tara whether a predecessor has been harmed, $\zeta_t^* = b$. Absent harm, the designer additionally informs Tara of the hypothetical outcomes ϕ_s of her neighbors in the original network G (but not of their actual outcomes ϕ_s^* induced by G^*). Since $\zeta_t^* = b$ perfectly reveals quality, the induced quality information is equivalent to revealing (ϕ_G, ζ_t^*) , which trivially Blackwell dominates ϕ_{G_t} and hence Pareto dominates G . The critical question is how the designer can learn the hypothetical outcomes ϕ_s . The simple solution is to ask Sven for his consumption $c_s = c(\phi_G, \theta_s)$ in the hypothetical scenario that he doesn't learn ζ_s^* ; absent incentives to misrepresent, we can assume that Sven reports c_s truthfully. The designer can deduce ϕ_s from the observed actual outcomes ϕ_s^* and c_s since, in the absence of harm, the new policy G^* makes agents more optimistic, induces more exploration, and thus provides the designer with weakly more information than the original network G .¹⁷

To implement the adjusted public warning G^* , the designer does not need to know quality q or agents' types θ , but must be able to (i) observe past outcomes, (ii) control agents' access to information, both by providing neighbors' hypothetical outcomes ϕ_s and by withholding their actual outcomes ϕ_s^* , and (iii) survey agents to elicit their hypothetical consumption c_t . The construction is "robust" (Wilson 1987) in that it achieves the Pareto improvement in any type space, and "detail free" (Dasgupta and Maskin 2000) in being formulated in terms of tangible choices instead of abstract types. However, the need to conceal information makes it unclear how to decentralize G^* .

Methodologically, the construction of the adjusted harm broadcast is ad hoc. A more principled approach would fix a canonical class of mechanisms $\mathcal{M}$ and ask if public warnings that reveal harm can Pareto improve outcomes within this class. To nest the network mechanisms from Section I, the messages m_t in $\mathcal{M}$ must be able to condition on past agents' outcomes, ϕ_s for $s < t$. But additionally, one needs to take a stance on whether they can condition on agents' types. The literature (e.g., Bergemann and Morris 2019) distinguishes between information design with known types, unknown types, and elicited types. Our social learning context creates a role for a novel, fourth, possibility, where the designer can condition Tara's message on the surveyed types θ_s of past agents $s < t$ but not on θ_t . Theorem 2 in Appendix A delivers a mixed message: Public warnings constitute Pareto improvements for mechanisms with known and surveyed types, but not for unknown or elicited types.

Public warnings are an intuitive idea and underlie important transparency policies. With binary outcomes, Theorem 1 provides a strong theoretical foundation: Public warning G^+ , adjusted public warning G^* , and full transparency $\bar{G}$ all coincide and are Pareto dominant for all types. With general outcomes, public warnings still constitute Pareto improvements under some informational assumptions.

¹⁷ *Proof:* To see how to learn ϕ_s from ϕ_s^* and c_s conditionally on $\zeta_s^* = n$, suppose inductively that the designer has learned the hypothetical outcomes ϕ_r for $r < s$. If $c_s = 0$, trivially $\phi_s = n$. If instead $c_s = 1$, then Sven also consumes when the additional information of no harm, $\zeta_s^* = n$, makes him more optimistic $c_s^* = c(\phi_G, \zeta_s^*, \theta_s) = 1$, and so the hypothetical outcome equals the actual outcome, $\phi_s = \varphi_s = \phi_s^*$.

III. Conclusion

This paper studied a model where agents learn about a new product when it harms their neighbors. While the guinea pig effect suggests a strong conflict of interest, we showed that full transparency is best for everyone and makes early movers the optimal guinea pigs for late movers.

This result provides a clear qualitative contrast between bad news learning and good news learning, where full transparency is suboptimal. To quantify this contrast, consider a line network $G_t = \{t-1\}$ and the uniform cost types from Example 1. Under bad news, any agent's learning is upper bounded by the value of observing one uninformed neighbor. By contrast, under good news, standard arguments imply that social learning becomes perfect for large t (e.g., Board and Meyer-ter-Vehn 2021). Intuitively, good news spreads more easily than bad news in sparse networks, so, as the network grows, the added value of information design remains large under bad news but vanishes under good news.

This contrast between good and bad news events has practical implications for transparency laws (e.g., Fung, Graham, and Weil 2007). Policymakers should focus on collecting and disseminating information about dangerous products with large uncertain downsides (e.g., products that may be lethal) rather than those with large uncertain upsides (e.g., products that may be revolutionary). For products with many unknown aspects (e.g., a new drug like Ozempic may cure various diseases but also have side effects), transparency policy should focus on the negative effects.

We have framed our model in terms of learning about product quality. Bad news learning also naturally arises when agents learn whether they are being monitored in a moral hazard model. For example, taxpayers learn whether a rule is enforced, employees learn whether their boss is paying attention, and financial firms learn whether a regulator's inspections are meaningful. To spell out the mapping, a taxpayer "tries the product" of tax evasion, a vigilant tax authority renders tax evasion "low quality," and the sentencing of a tax cheat constitutes "harm." So reinterpreted, the discussion about the product seller in Section IIA means that transparency minimizes expected tax evasion under a vigilant tax authority.

APPENDIX A. INFORMATION DESIGN

Here, we formalize the information design discussion from the end of Section IIB. We consider mechanisms M that send each agent t a deterministic message $m_t = m(\phi^{t-1}, \theta^t)$ based on the outcomes ϕ_s of agents $s < t$ and types θ_s of agents $s \leq t$.¹⁸ We consider five classes of progressively more powerful mechanisms:

- (i) *Networks*: The designer chooses a network G , and agents learn the outcomes of their neighbors, $\mathcal{M}_n = \{M : m_t = \phi_{G_t}\}$. This coincides with the observation structure in Section I, albeit with general outcomes.

¹⁸Standard information design would allow for stochastic messages that additionally depend on quality and noise terms $m_t = m(\phi^{t-1}, \theta^t, q, \epsilon_t)$. We rule out direct dependence on q to stay true to the notion that the designer must learn quality from agents' outcomes and avoid the trivial result that revealing q is first-best in our model of pure informational externalities. Insisting on deterministic messages avoids distinctions between our subjective probabilities $\Pr_{\theta_t}(q, \varphi, \theta_{-t})$ and objective probabilities about ϵ_t .

- (ii) *Unknown types*: The designer does not know agents' types but can release any information based on past agents' outcomes, $\mathcal{M}_u = \{M : m_t = m(\phi^{t-1})\}$. The public warning $m_t = (\phi_{G^*}^+, \zeta_t^+)$ is in this class.
- (iii) *Survey mechanisms*: The designer surveys past agents' types and can release any information based on past agents' outcomes and types, $\mathcal{M}_s = \{M : m_t = m(\phi^{t-1}, \theta^{t-1})\}$. The adjusted public warning $m_t = (\phi_{G^*}^*, \zeta_t^*)$ is in this class.
- (iv) *Elicited types*: The designer asks agent t to report her type $\hat{\theta}_t$ before sending her a message in a way that truth telling $\hat{\theta}_t = \theta_t$ is incentive compatible, $\mathcal{M}_e = \{M : m_t = m(\phi^{t-1}, \hat{\theta}^t) \text{ such that } \hat{\theta}_t = \theta_t\}$.
- (v) *Known types*: The designer can release any information based on past agents' outcomes and types and the current agent's type, $\mathcal{M}_k = \{M : m_t = m(\phi^{t-1}, \theta^t)\}$.

Each class gives the designer more power than the last:

$$\mathcal{M}_n \subset \mathcal{M}_u \subset \mathcal{M}_s \subset \mathcal{M}_e \subset \mathcal{M}_k.$$

As in Section I, assuming strict preferences, any mechanism M admits a unique equilibrium, and we Pareto rank mechanisms based on the outcomes of this equilibrium.

Returning to the question of whether full transparency about harm is optimal, we say that a class of mechanisms $\mathcal{M}$ satisfies the *Breaking Bad News Improvement Property* (BBN) if for any $M \in \mathcal{M}$ there exists a Pareto-improving $M^* \in \mathcal{M}$ that breaks bad news ζ_t ; that is, any agent t learns whenever any $s < t$ has been harmed. For example, the construction of the adjusted public warning G^* shows that any mechanism $M \in \mathcal{M}_n$ can be Pareto improved by some $M^* \in \mathcal{M}_s$. BBN requires this Pareto-improving construction more strongly for every $M \in \mathcal{M}_s$.

THEOREM 2: *Assume general, nonbinary outcomes.*¹⁹

- (i) $\mathcal{M}_n$ does not satisfy BBN.
- (ii) $\mathcal{M}_u$ does not satisfy BBN.
- (iii) $\mathcal{M}_s$ satisfies BBN.
- (iv) $\mathcal{M}_e$ does not satisfy BBN.
- (v) $\mathcal{M}_k$ satisfies BBN.

¹⁹With binary outcomes, the counterexamples for parts (i) and (ii) do not work, and BBN obtains also in those cases. In contrast, the counterexample for part (iv) works also for binary outcomes, so BBN fails. Parts (iii) and (v) are also unchanged. Moreover, full transparency about outcomes $\bar{G}$ is Pareto dominant in $\mathcal{M}_n, \mathcal{M}_u$, but not in $\mathcal{M}_s, \mathcal{M}_e, \mathcal{M}_k$ where the designer can release additional information about types that makes some types better off.

PROOF:

Part (i) follows because the only mechanism $M^* \in \mathcal{M}_n$ that breaks bad news is the complete network, which is Pareto dominant only when outcomes are binary.

Part (ii) essentially follows from Example 3. Consider observable actions, mechanism M associated with the two-link network $r \leftarrow s \leftarrow t$, and any mechanism $M^+ \in \mathcal{M}_u$ that breaks bad news and allows Sven to observe Ruby's outcome (a mechanism that does not allow Sven to observe Ruby's outcome cannot improve on M for many types of Sven). A type θ_t who knows (i) that Ruby does not try in M but may be induced to consume by the absence of harm in M^+ with no-harm outcome $\phi_r^+ = cn$ and (ii) that Sven tries with perfectly informative outcome in M but does not consume when observing $\phi_r^+ = n$ (which is negative news about Ruby's type in M^+) learns quality perfectly in M but imperfectly in M^+ .

For part (iii), fix any $M \in \mathcal{M}_s$. To construct $M^* \in \mathcal{M}_s$ that breaks bad news and improves on M , we follow the adjusted public warning. In particular, the designer tells any agent, Tara, whether harm has occurred; if it has not, he additionally sends her the original “hypothetical” message $m_t = m(\phi^{t-1}, \theta^{t-1})$. He can deduce the “hypothetical” outcomes ϕ^{t-1} because consumption is higher under the new mechanism, $c_t^* = c(m_t, \zeta_t^*, \theta_t) \geq c_t$ for $\zeta_t^* = n$, and he can deduce who consumes in the original mechanism $c_t = c(m_t, \theta_t)$ from the surveyed type θ_r .²⁰

For part (iv), we construct types and a mechanism $M \in \mathcal{M}_e$ that is not Pareto dominated by any $M^* \in \mathcal{M}_e$ that breaks bad news. Specifically, suppose that there are events $\{w, x, y, z, b\}$. Type θ_s 's beliefs over these events and quality are:

	w	x	y	z	b
$q = H$	1/12	1/12		2/12	
$q = L$	1/12		1/12		6/12

His indifference belief is $x^*(\theta_s) = 1/2 + \varepsilon$ for small $\varepsilon > 0$. Mechanism M lets Sven choose between signals $\{\{w, x\}, \{y, z, b\}\}$ and $\{\{w, y\}, \{x, z, b\}\}$. Since the latter is uninformative about quality, $\Pr_{\theta_s}(H|\{w, y\}) = \Pr_{\theta_s}(H|\{x, z, b\}) = 1/3$, he prefers the former signal, whose posterior beliefs, $\Pr_{\theta_s}(H|\{y, z, b\}) = 2/9$ and $\Pr_{\theta_s}(H|\{w, x\}) = 2/3$, sandwich the indifference belief. There is another type θ'_s for whom signal $\{\{w, y\}, \{x, z, b\}\}$ constitutes perfect learning.

Any mechanism $M^* \in \mathcal{M}_e$ that breaks bad news and makes θ'_s no worse off must offer Sven a message that refines $\{\{w, y\}, \{x, z\}, \{b\}\}$. This latter information structure allows θ_s to achieve his favorite decisions—namely, consume in $\{x, z\}$ but not in $\{w, y, b\}$. However, this means that θ_s no longer consumes in state $\{w\}$, which reduces the learning of some subsequent agents.²¹

²⁰To appreciate the power of surveying types, recall mechanism M associated with $r \leftarrow s \leftarrow t$ from part (ii). Mechanisms $M^+ \in \mathcal{M}_u$ that break bad news cannot Pareto improve on M : If Sven observes ϕ_r^+ , Tara may be worse off; if Sven doesn't observe ϕ_r^+ , Sven may be worse off. The Pareto-improving mechanism $M^* \in \mathcal{M}_s$ avoids this dichotomy by showing Sven Ruby's hypothetical outcome ϕ_r instead of the actual ϕ_r^+ .

²¹In contrast to the elicitation mechanism $M \in \mathcal{M}_e$, the survey mechanisms in part (iii) offer Sven one partition of $\{w, x, y, z, b\}$. By separating out b , all types of Sven become more optimistic in the other elements of the partition and increase their consumption. This construction fails for M where the constraint imposed by θ'_s prevents the designer from maintaining θ_s 's consumption in event w when breaking bad news (even though the designer can now elicit types).

Part (v) follows as in part (iii). The only difference is that Tara's message in the absence of harm now also depends on her type $m(\phi^{t-1}, \theta^t)$. Critically, the designer can still reconstruct the outcomes of the agents from the original mechanism.²² ■

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²²In contrast to part (iv), $M^* \in \mathcal{M}_k$ Pareto improves M by imposing $\{\{w, x\}, \{y, z\}, \{b\}\}$ on θ_s and $\{\{w, y\}, \{x, z\}, \{b\}\}$ on θ_s , which requires (unelicited) knowledge of types.